



CHEMISTRY CH.3 — ACID, BASES AND SALT — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Chlor-Alkali Process
2. Indicators: Natural, Synthetic & Olfactory
3. pH Scale & Common pH Values
4. Strong vs Weak Acids & Bases
5. Neutralization, Salt Types & pH
6. Common Salts: Preparation, Properties & Uses
7. Water of Crystallization, Efflorescence & Deliquescence
8. Acids & Bases in Nature, Daily Life & Industry
9. Salt Classification: Normal, Acidic, Basic, Double, Complex & Mixed
10. Alums, Amphoteric Oxides & Water Purification
11. Soap vs Detergent & Water Softening
12. Acid Rain, Tooth Decay & Antacids
13. Metal Carbonates & Bicarbonates with Acids
14. ⚠ Traps / Key Differences Summary (Q275–Q447)

1. Chlor-Alkali Process

- Chlor-alkali process = aqueous NaCl ka electrolysis; products: Hydrogen (H₂), Chlorine (Cl₂), Sodium hydroxide (NaOH). [Q275, Q277]
- Reaction: $2\text{NaCl(aq)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$. [Q275]
- Use: soap, detergents, paper, PVC, bleach manufacturing mein. [Q275]

2. Indicators: Natural, Synthetic & Olfactory

- Natural indicators: Red cabbage leaves (purple→red in acid, green/yellow in base), turmeric, litmus. [Q276, Q278]
- Synthetic indicators: Methyl orange, phenolphthalein, universal indicator. [Q276, Q278]
- Olfactory indicators = jinki smell acidic ya alkaline medium mein change ho jaati hai: Onion, clove, vanilla — acid mein apni smell lose karte hain, base mein retain karte hain. [Q276, Q280, Q359, Q369, Q381]
- Red cabbage leaves = natural acid-base indicator. [Q276]
- Methyl orange = synthetic acid-base indicator; basic solution mein yellow ho jaata hai. [Q276, Q336]
- Onion, clove = olfactory indicator. [Q276]
- Litmus solution = purple dye jo lichen (ek plant) se extract kiya jaata hai; neutral solution mein iska colour purple rehta hai. [Q315, Q319, Q367, Q385]
- Jasmine flower leaves acid/base presence indicate nahi karte — ye indicator nahi hain. [Q386]
- Statement check: Olfactory indicator ek material hai jiski smell acid/base mein change hoti hai — ye statement sahi hai. [Q285]
- Iodine solution starch ki presence detect karne ke liye use hota hai (starch-specific indicator, acid-base nahi). [Q307]
- NaOH (sodium hydroxide) solution pH paper ko blue kar deti hai (basic nature ki wajah se). [Q311]
- Borax ek basic substance hai — litmus add karne par ye blue ho jaata hai. [Q332]
- Phenolphthalein alkalis mein pink ho jaata hai; acids mein colourless rehta hai. [Q347]

- Sulphuric acid (H_2SO_4) blue litmus ko red kar deta hai (acidic nature). [Q413]
- Dry HCl gas dry litmus paper ka colour change nahi karta — ionization ke liye water zaroori hai. [Q423]
- Universal indicator ek mixture hai jo different pH par different colours deta hai — exact pH measure karne ke liye best indicator hai; ye "metals aur non-metals ka mixture" nahi hota (galat statement). [Q415, Q432]
- Phenolphthalein acidic aur neutral solutions ke beech distinguish nahi kar sakta — dono mein colourless rehta hai. [Q428]

3. pH Scale & Common pH Values

- pH scale = 0 se 14; pH 7 = neutral (pure water at 25°C). [Q279, Q281, Q283, Q397]
- pH < 7 = acidic; pH > 7 = basic/alkaline. [Q279, Q281]
- Strong acid: pH 0–3 (HCl , H_2SO_4 , HNO_3); Weak acid: pH 4–6 (acetic acid, citric acid). [Q283]
- Strong base: pH 11–14 (NaOH , KOH); Weak base: pH 8–10 (NH_4OH). [Q283]
- Lemon juice: pH $\approx$ 2.0–2.5 — solutions mein sabse minimum pH value in hone ke examples mein se ek. [Q391, Q393, Q395, Q398]
- Vinegar: pH $\approx$ 2.4–3.4; Black coffee: pH $\approx$ 5.0. [Q391, Q393, Q395]
- Milk: pH $\approx$ 6.6; Pure water: pH = 7.0; Blood: pH $\approx$ 7.35–7.45. [Q395, Q397, Q399]
- Human body ka normal pH range approximately 7.0 se 7.8 ke beech hota hai. [Q438]
- Baking soda solution: pH $\approx$ 8.3; Milk of magnesia: pH $\approx$ 10.5; Drain cleaner: pH $\approx$ 13–14. [Q399, Q401, Q403]
- $\text{pH} = -\log[\text{H}^+]$; $[\text{H}^+] = 10^{-\text{pH}}$. [Q287]
- $[\text{H}^+] = 1 \times 10^{-5} \text{ mol/L}$ ho to pH = 5 hoga. [Q425]
- Universal indicator mein red colour pH 0 se 3 ko show karta hai (highly acidic). [Q339]
- Highly acidic mixture ka pH value 0 ke kareeb hoti hai. [Q390]
- Acid strong ionize karta hai to zyada H^+ ions produce hote hain, jitni zyada concentration utni zyada acidity — is se solution zyada acidic ban jaata hai jaise pH = 4 wala solution pH = 7 se zyada acidic hoga (zyada H^+ concentration). [Q301, Q329, Q355]
- High concentration of hydroxide (OH^-) ions wala solution basic hota hai aur uska pH value high hoti hai. [Q309]
- Neutral solution ka pH value exactly 7 hota hai — na acidic na basic. [Q371]
- Blue litmus red ho jaaye to solution ka pH 7 se kam hoga (acidic). [Q326]
- Red litmus blue ho jaaye to solution basic hoga, jaise pH $\approx$ 10 wala solution. [Q417]
- pH badhne ke saath (7 se 14 tak) OH^- ion concentration increase hoti hai. [Q361]
- Pure water mein H^+ aur OH^- ions ki concentration barabar (equal) hoti hai — isiliye ye neutral hai. [Q375]
- pH 3 se pH 6 hone par H^+ ion concentration 1000 guna kam ho jaati hai (har unit pH change = 10x factor). [Q421]
- Alkaline solution mein $[\text{H}^+] < 1 \times 10^{-7} \text{ mol/L}$ hoti hai. [Q434]
- Concentrated acid ko dilute karne ke liye hamesha acid ko water mein add karte hain, water ko acid mein nahi (safety ke liye — exothermic reaction). [Q334, Q440]
- Light rain (pH < 5.6) jab river mein jaata hai to river water ka pH decrease ho jaata hai. [Q317]
- ⚠ pH 7 neutral sirf 25°C (298 K) par hota hai; temperature change se water ka pH badalta hai. [Q397]

4. Strong vs Weak Acids & Bases

- Strong acids = water mein completely dissociate: HCl , H_2SO_4 , HNO_3 , HBr , HI , HClO_4 . [Q282, Q284, Q286]
- Weak acids = partially dissociate: CH_3COOH (acetic acid/vinegar), H_2CO_3 (carbonic acid), citric acid, tartaric acid, oxalic acid. [Q284, Q286, Q370, Q372]

- Strong bases = completely dissociate: NaOH (caustic soda), KOH (caustic potash), Ba(OH)₂. [Q288, Q290, Q292]
- Weak bases = partially dissociate: NH₄OH (ammonia solution), Mg(OH)₂, Al(OH)₃. [Q288, Q290]
- Ca(OH)₂ = slaked lime (strong base, slightly soluble); CaO = quick lime; CaCO₃ = limestone. [Q292]
- Listed weak acids mein sabse weak acetic acid ya benzoic acid ho sakta hai depending on comparison — dono weak acids ke examples hain. [Q321, Q363, Q400]
- Acidic solutions water mein electricity conduct karte hain kyunki ye H⁺ (hydrogen) ions produce karte hain. [Q341, Q402]
- Acid mein zyada ionize hokar zyada H⁺ dena "stronger acid" banata hai; base bhi similarly water mein completely ionize hoke zyada OH⁻ deti hai to "stronger base" banti hai. [Q405, Q407]
- Water solution mein: Acids H⁺ dete hain; Bases OH⁻ dete hain. [Q409]
- Metals acids ke saath react karke salt aur Hydrogen gas dete hain. [Q419]
- Weak acid aur uske salt ka solution buffer solution kehlata hai — ye pH change ko resist karta hai. [Q327]
- Acids ki common properties: sour taste, blue litmus ko red kar dete hain. [Q351]
- Alkalis (bases) ka taste sour nahi hota — ye bitter/soapy hota hai; sour taste acids ki property hai, alkalis ki nahi. [Q387]

5. Neutralization, Salt Types & pH

- Neutralization = Acid + Base → Salt + Water + Heat. [Q289, Q291, Q293]
- Example: HCl + NaOH → NaCl + H₂O; H₂SO₄ + 2NaOH → Na₂SO₄ + 2H₂O. [Q289, Q291]
- Neutralization exothermic reaction hai; neutral salt ka pH = 7. [Q293]
- Strong acid + Strong base → Neutral salt (pH 7); jaise NaCl (strong acid HCl + strong base NaOH). [Q295, Q389, Q436]
- Strong acid + Weak base → Acidic salt (pH < 7); zero ke kareeb pH wala salt strong acid + weak base se banta hai. [Q295, Q392]
- Weak acid + Strong base → Basic salt (pH > 7); jaise Sodium carbonate (weak carbonic acid + strong NaOH). [Q295, Q349]
- Normal salt = acid ka all replaceable H⁺ replace ho: NaCl, Na₂SO₄. [Q427, Q429]
- Acid salt = partial H⁺ replace ho: NaHSO₄ (sodium bisulphate), NaHCO₃. [Q429, Q431]
- Basic salt = partial OH⁻ replace ho: Cu(OH)Cl (basic copper chloride). [Q431, Q433]
- Ammonium phosphate ek acidic salt ka example hai. [Q430]

6. Common Salts: Preparation, Properties & Uses

- Sodium chloride (NaCl) = common salt; food preservative, chlor-alkali raw material; molecular weight ≈ 58 g/mol (Na=23 + Cl=35.5). [Q294, Q296, Q322]
- Sodium bicarbonate (NaHCO₃) = baking soda; mild base; baking, antacid, fire extinguisher. [Q296, Q298, Q338, Q340, Q342]
- Preparation: NaCl + NH₃ + CO₂ + H₂O → NaHCO₃ + NH₄Cl (Solvay process). [Q338]
- Heating: 2NaHCO₃ → Na₂CO₃ + H₂O + CO₂. [Q342]
- Sodium carbonate (Na₂CO₃·10H₂O) = washing soda; laundry, glass, paper, water softening industry mein use; iske crystals mein 10 water of crystallisation molecules hote hain. [Q296, Q298, Q344, Q346, Q348, Q299]
- Baking soda se heat karke washing soda banata hai; phir crystallization. [Q344]
- Washing soda hygroscopic hai; efflorescence dikha sakta hai. [Q348]
- Bleaching powder (CaOCl₂) = calcium oxychloride; disinfectant, bleaching, water treatment. [Q298, Q300, Q331, Q333, Q335]

- Manufacturing: $\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$ (Back process). [Q331, Q333]
- Acid ke saath: $\text{CaOCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{Cl}_2 + \text{H}_2\text{O}$. [Q337]
- Plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$) = surgery, decorative work, toys, fracture setting. [Q300, Q302, Q350, Q352, Q354]
- Preparation: Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) ko 373 K par heat karke. [Q350, Q352]
- Water add karne par hard crystalline mass banata hai (expansion hota hai). [Q354]
- Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) = cement manufacturing mein setting time control. [Q356, Q358]

7. Water of Crystallization, Efflorescence & Deliquescence

- Water of crystallization = salt crystals mein fixed number of chemically bound water molecules. [Q360, Q362, Q364]
- Examples: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Blue vitriol, 5 water); $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (Green vitriol, 7 water); $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (10 water). [Q360, Q362, Q364]
- Efflorescence = hydrated salt air mein water lose karke powdery ban jaata hai. Washing soda: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O} + 9\text{H}_2\text{O}$. [Q364, Q366]
- Deliquescence = solid air se moisture absorb karke solution ban jaata hai. Example: CaCl_2 , NaOH , KOH . [Q366, Q368]
- Hygroscopic = moisture absorb karte hain bina dissolve hue. Example: Conc. H_2SO_4 , silica gel. [Q368]

8. Acids & Bases in Nature, Daily Life & Industry

- Citric acid = citrus fruits; Acetic acid (CH_3COOH) = vinegar (5–8% concentration); Tartaric acid = tamarind, grapes. [Q370, Q372, Q374, Q357]
- Lactic acid = curd, milk; muscle cramps during exercise; ek three-carbon molecule hai. [Q374, Q376, Q324]
- Oxalic acid = spinach, tomatoes; Formic acid (HCOOH /Methanoic acid) = ant/nettle sting. [Q376, Q378, Q380, Q383, Q396]
- Ant bite se hone wala acidic effect moist baking soda ya Calamine solution rubbing karke neutralize kiya ja sakta hai. [Q323]
- Ascorbic acid (Vitamin C) = citrus, amla; scurvy prevention — "Ascorbic acid - Tamarind" ek incorrect pairing hai (tamarind mein tartaric acid hota hai, ascorbic acid nahi). [Q380, Q382, Q330]
- Boric acid eyewash solutions mein constituent ke roop mein use hota hai. [Q325]
- Muriatic acid, Hydrochloric acid (HCl) ka hi scientific/commercial naam hai. [Q328]
- Baking powder = Sodium hydrogen carbonate (NaHCO_3) + Tartaric acid ka mixture hota hai. [Q353, Q373]
- Sulphuric acid (H_2SO_4) = "King of chemicals"; fertilizers, dyes, paints, detergents, petroleum refining; ise water mein dissolve karna highly exothermic hota hai. [Q404, Q406, Q408, Q384]
- Nitric acid (HNO_3) = fertilizers, explosives (TNT), dyes. [Q406, Q408]
- Hydrochloric acid (HCl) = steel cleaning, pickling, food processing. [Q408, Q410]
- Phosphoric acid (H_3PO_4) = soft drinks, fertilizers, dental cement. [Q412, Q414]
- Sodium hydroxide (NaOH) = soap, detergents, paper, textiles, drain cleaner; basic nature ka strong example. [Q416, Q418, Q420, Q345]
- Calcium hydroxide (Ca(OH)_2) = whitewash, soil treatment, sugar refining. [Q418, Q420]
- Ammonia (NH_3) = fertilizers, nitric acid, refrigeration, cleaning. [Q420, Q422]
- Magnesium hydroxide (Mg(OH)_2) = milk of magnesia; antacid, laxative. [Q422, Q424]
- Aluminium hydroxide (Al(OH)_3) = antacid, water purification, dyeing mordant. [Q424, Q426]
- Hydrogen element sabhi acids mein common hota hai (acid ki defining property). [Q379]
- CCl_4 (Carbon tetrachloride) acidic nahi hota hai. [Q377]

- Ester generally ek acid aur alcohol ki reaction se banta hai. [Q388]

9. Salt Classification: Normal, Acidic, Basic, Double, Complex & Mixed

- Double salt = do simple salts equimolar proportions mein crystallize; Mohr's salt ($\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$), potash alum. [Q433, Q435]
- Complex salt = complex ion banta hai; $\text{K}_4[\text{Fe}(\text{CN})_6]$ (Potassium ferrocyanide). [Q435, Q437]
- Mixed salt = ek salt mein do different anions ya cations; CaOCl_2 (bleaching powder), NaKCO_3 . [Q437, Q439]

10. Alums, Amphoteric Oxides & Water Purification

- Alum = double sulphate salt; general formula $\text{M}^+\text{M}^{3+}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$. [Q439, Q441, Q443]
- Potash alum = $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$; water purification, dyeing, baking powder. [Q441]
- Chrome alum = $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$; tanning, photography. [Q443]
- Alum water purification mein coagulating agent ki tarah kaam karta hai — suspended particles settle ho jaate hain. [Q441, Q443]
- Metal oxides generally basic nature ke hote hain — acid ke saath react karke salt + water banate hain (jaise Na_2O , MgO , CaO). [Q305, Q313]
- $\text{Cu}(\text{OH})_2$ ek alkali nahi hai — kuch metal hydroxides insoluble hote hain aur alkali category mein nahi aate. [Q297]
- Al_2O_3 (Aluminium oxide) ek amphoteric oxide hai — ye acids aur bases donon ke saath react karta hai, isliye "basic metal oxides" ke general rule ka odd-one-out example hai. [Q365, Q394, Q442]

11. Soap vs Detergent & Water Softening

- Soap = sodium/potassium salt of long-chain fatty acids; animal fats/vegetable oils ka NaOH/KOH se hydrolysis (saponification). [Q444, Q445, Q446]
- Synthetic detergent = sodium salts of long-chain benzene sulphonic acids; petroleum se bante hain. [Q444, Q445]
- Soap hard water mein achchha kaam nahi karta kyunki $\text{Ca}^{2+}/\text{Mg}^{2+}$ se insoluble scum (calcium stearate) banata hai. [Q446, Q447]
- Detergent hard water mein bhi kaam karta hai; $\text{Ca}^{2+}/\text{Mg}^{2+}$ se soluble complex nahi banata. [Q446, Q447]
- Soap biodegradable hai; kuch detergents non-biodegradable hain (environmental pollution). [Q447]
- Soap cleansing micelle formation se hoti hai — hydrophobic tail grease mein, hydrophilic head water mein. [Q445, Q447]
- Temporary hardness = $\text{Ca}(\text{HCO}_3)_2/\text{Mg}(\text{HCO}_3)_2$; boiling se remove: $\text{Ca}(\text{HCO}_3)_2 \rightarrow \text{CaCO}_3 \downarrow + \text{H}_2\text{O} + \text{CO}_2$. [Q346]
- Permanent hardness = $\text{CaSO}_4/\text{CaCl}_2$; washing soda (Na_2CO_3) se remove: $\text{CaSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 \downarrow + \text{Na}_2\text{SO}_4$. [Q346, Q348]
- Zeolite/Permutit process = ion exchange se water softening. [Q348]

12. Acid Rain, Tooth Decay & Antacids

- Acid rain = $\text{pH} < 5.6$; H_2SO_4 aur HNO_3 se banta hai. [Q304, Q306, Q308]
- Causes: SO_2 (fossil fuels) aur NO_x /Nitrogen oxides (vehicles) emission — sirf SO_2 nahi, nitrogen oxides bhi acid rain cause karte hain. [Q304, Q306, Q343]
- Effects: marble/statues corrode ($\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$), aquatic life damage, soil infertility. [Q306,

Q308]

- Taj Mahal ka yellowing = acid rain effect. [Q308]
- Tooth enamel = calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$); hardest body substance. [Q310, Q312, Q411]
- $\text{pH} < 5.5$ par enamel corrode hoti hai; bacteria acid produce karte hain. [Q310]
- Toothpaste basic hota hai; acid neutralize karke enamel protect karta hai; fluoride enamel strengthen karta hai. [Q312]
- Antacids = excess stomach acid neutralize karne ke liye mild bases hain — ye strong acids nahi hote (common misconception). [Q314, Q316, Q318, Q303]
- Examples: $\text{Mg}(\text{OH})_2$ (Milk of magnesia), $\text{Al}(\text{OH})_3$, NaHCO_3 (Baking soda — antacid ka active ingredient), CaCO_3 . [Q314, Q316, Q320]
- Strong base (NaOH , KOH) antacid ke roop mein nahi use hote — harmful hain. [Q316]
- Ranitidine, Omeprazole = modern antacids; acid production kam karte hain. [Q318]

13. Metal Carbonates & Bicarbonates with Acids

- Metal carbonates + dilute acid $\rightarrow$ salt + water + CO_2 . [Q340]
- $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$. [Q346]
- $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$ (baking soda antacid action). [Q340]
- CO_2 gas limewater ($\text{Ca}(\text{OH})_2$) ko milky (CaCO_3 precipitate) banati hai — CO_2 test. [Q337]

14. ⚠ Traps / Key Differences Summary (Q275–Q447)

- ⚠ Trap: Chlor-alkali process mein Oxygen nahi, $\text{H}_2 + \text{Cl}_2 + \text{NaOH}$ products hain. [Q275]
- ⚠ Trap: Olfactory indicators = onion/clove/vanilla; Natural = red cabbage/turmeric/litmus; Synthetic = methyl orange/phenolphthalein — alag categories hain. [Q276]
- ⚠ Trap: Litmus lichen (plant) se extract hota hai — ye ek natural indicator hai jiska neutral colour purple hota hai. [Q315, Q319, Q367, Q385]
- ⚠ Trap: $\text{pH} 7$ neutral sirf 25°C par hota hai; garam pani ka $\text{pH} 7$ se kam hota hai lekin still neutral. [Q397]
- ⚠ Trap: $\text{Ca}(\text{OH})_2$ = slaked lime (strong base); CaO = quick lime; CaCO_3 = limestone — teenon alag hain. [Q292]
- ⚠ Trap: Caustic soda = NaOH ; Caustic potash = KOH ; Washing soda = Na_2CO_3 — caustic vs washing confuse mat hona. [Q292, Q346]
- ⚠ Trap: Neutralization hamesha salt + water deta hai; CO_2 nahi (carbonate reaction alag hai). [Q289]
- ⚠ Trap: Strong acid + Weak base $\rightarrow$ acidic salt; Weak acid + Strong base $\rightarrow$ basic salt — final pH decide karta hai kaunsa strong hai. [Q295]
- ⚠ Trap: Baking soda = NaHCO_3 ; Washing soda = $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$; Baking powder = $\text{NaHCO}_3 + \text{Tartaric acid}$ — teenon alag. [Q338, Q344, Q353]
- ⚠ Trap: Plaster of Paris $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$; Anhydrite CaSO_4 — alag hydration states. [Q350, Q356]
- ⚠ Trap: Efflorescence = water lose karke powdery; Deliquescence = moisture absorb karke liquid — opposite. [Q364, Q366]
- ⚠ Trap: Hygroscopic = moisture absorb bina dissolve hue; Deliquescent = dissolve ho jaata hai — alag hain. [Q366, Q368]
- ⚠ Trap: Bleaching powder formula CaOCl_2 hai; CaCl_2 nahi — chlorine oxidation state alag hai. [Q331, Q333]
- ⚠ Trap: Formic (Methanoic) acid ant sting mein; Acetic acid vinegar mein — donon alag organic acids. [Q378, Q380, Q396]
- ⚠ Trap: Tooth decay $\text{pH} < 5.5$ par shuru hoti hai; normal mouth $\text{pH} \approx 6.5\text{--}7.5$; body ka overall pH range $7.0\text{--}7.8$

hai. [Q310, Q438]

- ⚠ Trap: NaOH aur KOH strong base hain antacid ke roop mein nahi use hote — internal tissues damage kar sakte hain. [Q316]
- ⚠ Trap: Sulphuric acid "King of chemicals" hai kyunki sabse zyada use hota hai — iska matlab ye nahi ki sabse strong hai. [Q404, Q406]
- ⚠ Trap: Acid salt mein replaceable H^+ bacha rehta hai ($NaHSO_4$); Normal salt mein sab H^+ replace ho gaya (Na_2SO_4). [Q429, Q431]
- ⚠ Trap: Double salt crystals mein equimolar salts hain; Complex salt mein complex ion hota hai — donon alag hain. [Q433, Q435]
- ⚠ Trap: Alum ka general formula $M^+M^{3+}(SO_4)_2 \cdot 12H_2O$ hota hai; sabhi alums sulphate hain. [Q439, Q441]
- ⚠ Trap: Soap hard water mein scum banata hai; detergent nahi banata — isliye detergent hard water mein better hai. [Q446, Q447]
- ⚠ Trap: Temporary hardness boiling se jaati hai; Permanent hardness washing soda/ion exchange se jaati hai — alag methods. [Q346, Q348]
- ⚠ Trap: Acid rain mein H_2SO_4 aur HNO_3 donon hote hain; sirf SO_2 nahi, nitrogen oxides bhi cause karte hain. [Q306, Q343]
- ⚠ Trap: Acid rain ka $pH < 5.6$ hota hai; normal rain ka $pH \approx 5.6$ (CO_2 ki wajah se slightly acidic). [Q304]
- ⚠ Trap: Concentrated acid dilute karte waqt hamesha acid ko water mein daalo, ulta karne se violent exothermic reaction ho sakti hai. [Q334, Q440]
- ⚠ Trap: Al_2O_3 amphoteric hai — sirf basic metal oxides ke general rule ka exception hai. [Q365, Q394, Q442]
- ⚠ Trap: Phenolphthalein acidic aur neutral solutions mein colourless rehta hai — inhe distinguish nahi kar sakta; alkalis mein pink hota hai. [Q347, Q428]